SUPPORTING ADULT GUIDANCE

Growth models

Percentages, Ratio, Growth and Bounds · Lesson 7. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The lesson opens with two ways a message spreads (100 people plus 10 a day, or 1 person doubling each day). Movement sorts sequences into add, multiply or neither, with the differences and ratios shown. Reflection reads rules and tables, with a stretch on why multiplying overtakes adding. The main activity, the Model Builder, is in Creativity: learners build two models, each set to + add or × multiply, with a start and a step. Both tables (with difference and ratio columns) and both graphs, on one grid with numbered, labelled axes, update together. The readout names where the lead changes and rings each overtake on the graph. Every value is exact: a ratio that is not a terminating decimal is shown as a fraction. Six briefs ask learners to match tables, model a phone contract, set up an overtake at a chosen step, and make two models meet twice.

Driving Question: *Add the same, or multiply by the same?*

The mathematics, in plain terms

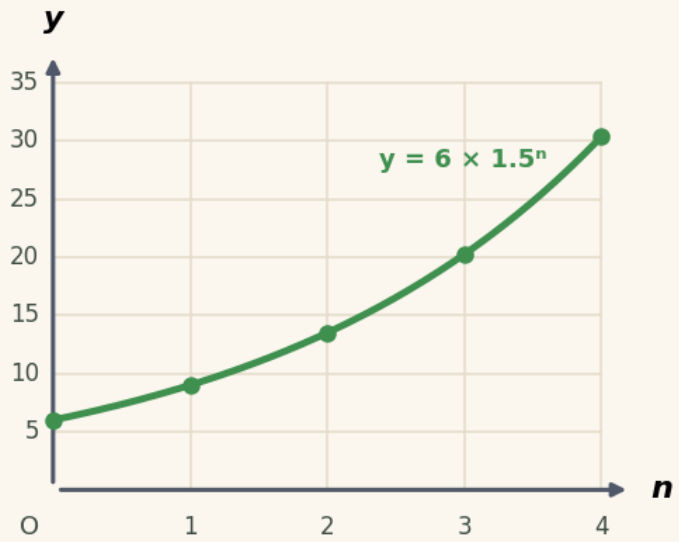
Linear: $y = a + dn$, first differences constant. Exponential: $y = am^n$, ratios constant. The exponential model's step is $a(m - 1)m^{n-1}$, which grows without limit for $m > 1$, so it eventually exceeds any constant d and the exponential model overtakes. Because the exponential curve is convex, a line can meet it at most twice; between those points the line is above the curve.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Which model fits?

A table goes 6, 9, 13.5, 20.25, … Is it linear or exponential? Find its rule and the next value.

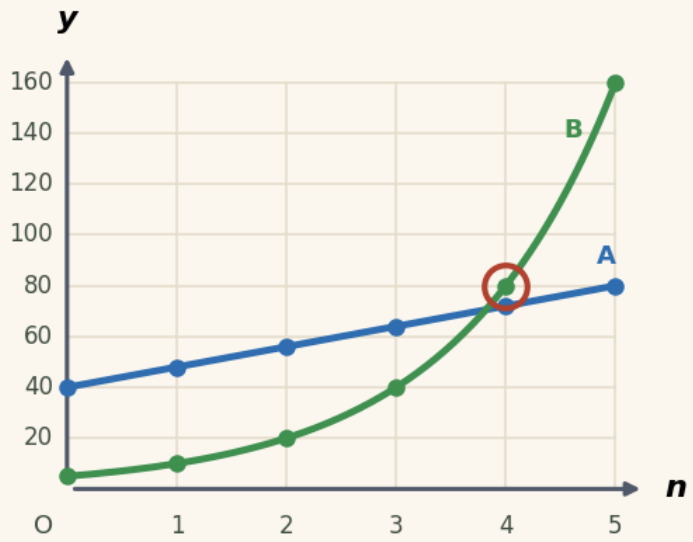


1.	The differences	3, 4.5, 6.75: not the same
2.	The ratios	$\frac{9}{6} = 1.5$, $\frac{13.5}{9} = 1.5$, $\frac{20.25}{13.5} = 1.5$
3.	The rule	$y = 6 \times 1.5^n$
4.	The next value	$20.25 \times 1.5 = 30.375$

Exponential growth, $y = 6 \times 1.5^n$; the next value is 30.375.

Example 2 — When does it overtake?

Model A is $y = 40 + 8n$. Model B is $y = 5 \times 2^n$. At which n does B first overtake A?



1. $n = 2$	A: 40 + 16 = 56; B: 5 × 4 = 20
2. $n = 3$	A: 64; B: 40
3. $n = 4$	A: 72; B: 80

B first overtakes A at $n = 4$, and pulls further ahead every step after that.

Build, then test, then choose

Learners sort sequences by their differences and ratios before writing rules, then build models to match given tables and stories. Seeing both tables and both graphs change together lets them connect “constant difference” with a straight line and “constant ratio” with a curve.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Two ways to spread a message; the idea card.	~5 min
Movement	Add or multiply? Eight sequences.	~6 min
Reflection	Reading rules and tables; the overtake stretch.	~10 min
Creativity	The Model Builder (main activity) and six briefs.	~19 min
Rest	Steady growth and growth that builds on itself.	~2 min
Nutrition	Practical: reading claims about growth.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Calling any increasing sequence linear.
Ask: “Are the differences all the same?”

Applying the power to the start value: 3×2^5 read as 6^5.
Ask: “Which number is multiplied each step?”

Using the increase as the multiplier (0.25 for a 25% increase).
Ask: “What do you multiply by to get the new value?”

Thinking a model that starts ahead stays ahead.
Ask: “What happens to each model’s step as n grows?”

Mixing up *a* and *d* in $y = a + dn$.
Ask: “What is y when $n = 0$?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: straight-line graphs, sequences and repeated change (Lesson 6). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says 5, 15, 45, 135 is linear because it keeps going up. The aim is a question that sends them to the differences.

- “What are the differences?”
- “Are they the same?”

- “What do you multiply 5 by to get 15? And 15 to get 45?”
- “So which kind of model is it?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why do some ratios show as fractions?

So every value is exact. A linear model's ratios are often not terminating decimals (for example 25 over 23).

Why can't the adding model go below 0?

The graph shows the first quadrant only, so the builder keeps both models at or above 0 over the steps shown.

Is the stretch required?

No. The secure core is telling linear from exponential by differences and ratios, and writing the rule.

What comes next?

Lesson 8: bounds of a rounded value, asking what a number could have been before rounding.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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